DESCENT OF REGULARITY ALONG A SPLIT EXACT CONORMAL SEQUENCE FOR IMMERSIONS OF SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10672-10688 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 10690-10693, the complete author proof (2 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Companion to this wave's L16189/L16190 and to wave 14's L16129 (quasi-regular transitivity, B side).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma transitivity-conormal (in-source on this side; schemes twin carded as S3-morph-transitivity-conormal-schemes); Divisors, lemma divisors-lemma-permanence-regular-immersion.

Source reference. Original lemma 068Z: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10693–10693. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06AE: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 591–603. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $i : Z \to Y$ and $j : Y \to X$ be immersions of algebraic spaces over S. Assume j is locally of finite presentation and that the sequence*

$$0 \to i^*\mathcal{C}_{Y/X} \to \mathcal{C}_{Z/X} \to \mathcal{C}_{Z/Y} \to 0$$

of Lemma 06AE is exact and locally split.

(1) *If $j \circ i$ is a quasi-regular immersion, so is i.*
(2) *If $j \circ i$ is a H_1-regular immersion, so is i.*
(3) *If both j and $j \circ i$ are Koszul-regular immersions, so is i.*

Proof. Immediate from the case of schemes, see Divisors, Lemma 068Z. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.